

LPM0 turns off the CPU and leaves everything else functional. LPM1 and LPM2 add various clocking functions to the list of disabled functions.

LPM3 is the most used LPM, leaving only a low-frequency clock oscillator—and any peripheral that uses that clock—running. LPM3 is often called the RTC mode because it allows a timer to operate from a low-power 32768Hz clock source, consuming less than 1 μ A, and periodically wakes up the system for activity.

Finally, LPM4 turns off all clocks on the device, thus turning off any peripheral that uses clocks automatically. Analogue peripherals may still be active. If none of them is active, current consumption of LPM4 is only 100 nA, including RAM-retention. Any enabled interrupt can bring the MCU out of LPMs.

MSP430 doesn't support 1.8V operation, but Texas Instruments' latest addition MSP430F5xx operates at 12 MHz with 1.8V supplied. It specifies pin leakage currents of 50 nA to 1 μ A maximum.

Although zero-power BOD used in MSP430 can save a lot of power, it is extremely slow and can require a millisecond to detect a below-threshold voltage. Slow response could put the controller at risk. In addition, detection level is only 1.4V—substantially lower than the 2.2V required for having all the MCU features available. Brown-out protection of MSP430 cannot be disabled.

The analogue-to-digital converter (ADC) of MSP430 is an example of intelligent peripherals. If it is not actively converting, it draws no current. It automatically disables some of its internal oscillators and digital circuitry when it is not converting, effectively turning itself off automatically. The required circuitry is then re-enabled, or turned on, automatically if a conversion is triggered. Again, the CPU does not have to wake up to initiate the ADC conversion. The timer controls the timing and initiates the conversion. The

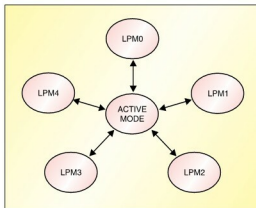


Fig. 9: MSP430 CPU mode transitions

ADC can then wake up the CPU after conversions complete.

A few MSP430 MCUs having DMA capability automatically handle data without CPU intervention. Using a DMA controller can increase the data-handling throughput, but, more importantly, it lowers power consumption. Using DMA automatically moves ADC data to RAM and allows the CPU to remain sleeping while ADC conversions take place. It wakes up the CPU only when all the desired conversions are complete.

Peripheral clock control is another feature that reduces current consumption of the system. The I²C module of MSP430 has direct control over the system clock. This means if the I²C module needs a clock, it automatically enables it. This again allows the CPU to remain asleep longer, which provides the ability to catch the incoming I²C address without having to wake up the CPU to enable the I²C module. MSP430 provides interrupts for 16 input/output (I/O) pins and all peripherals. Some peripherals have multiple interrupts for total flexibility.

MSP430 allows for a clean atomic transition to any of the five supported LPMs because the bits controlling the various clock domains and the general interrupt-enable bit are all located in the same CPU status register (SR). While atomic transition to any LPM is natural in MSP430, each ISR must explicitly disable the LPM in the

stacked SR, so that the CPU doesn't automatically return to the LPM, but rather the background loop can continue after the ISR restores the timer keeps running. When waking up either from power-down or power-save mode, there is a delay until the wake-up becomes effective. It is usually less than 1 μ s when using the internal RC oscillator.

Standby mode is identical to power-down and extended standby mode is identical to power-save mode with the exception that the crystal oscillator runs to ensure the shortest possible start-up time. The MCU wakes up in six clock cycles from either mode.

external triggered interrupts as well as internal ones.

Power-down mode halts all generated clocks, allowing operation of only asynchronous modules, whereas in power-save mode the timer keeps running. When waking up either from power-down or power-save mode, there is a delay until the wake-up becomes effective. It is usually less than 1 μ s when using the internal RC oscillator.

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Typical Current Consumption in Different Modes at 1.8V/1 MHz

Condition	Value
1.8V Idle mode	40 μ A
1.8V Power-save mode	4.5 μ A
1.8V Power-down mode	100 nA

AVR supports a true 1.8V operation, which means that all the analogue modules, Flash, EEPROM and RAM run at 1.8V. True 1.8V operation ensures less power consumption as it is the sum of current and voltage allowing the battery supply to drop to 1.8V before being out of range.

The leakage current of picoPower AVRs is less than 100 nA.